SUPPLEMENT: PREDICTING EXECUTION PATHS AND BOUNDING OPTIMIZATION IN SPECULATIVE DECODING

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A SUPPLEMENT SCOPE

This separate appendix accompanies the prospective prediction, sampled intervention, qualification/cost and runtime studies. It supplies statistical, measurement-boundary, and provenance details. Section J and its commitment tables document the main B570 H1 result. Section L and the post-study tables cover the new forks; Section M and the input audit cover RTX prediction; Section N covers RTX cost. M4 appears in Section H and the post-study fixed-mode comparison; 358H appears in Section D and the conditional accounting. Section O identifies document and earlier study packages; Section Q gives RTX V4 same-set prediction, matched cost, failed V3 history and full anonymous evidence entries. Identifying repository metadata and internal source paths remain outside the anonymous PDF; the nonidentifying B570 prediction-file content digest appears below.

B FROZEN ESTIMATORS

Table A1 preserves historical hardware estimators only; M4 runtime is in Section H, new RTX cost in Section N, and new post-study sensitivity in Section P. The primary sample unit is the prompt. Repeated observations contribute to its summary, rather than being treated as independent prompts. Pair ratios divide matched baseline decode time by speculative decode time. Historical M4, RTX, and B570 counts are text-path agreement from parsed CLI output: M4 HumanEval includes the prompt in its parsed text; B570 strips and normalizes the prompt before re-tokenization. Neither is direct generated-ID equality. The 358H runner and prospective M4 study record generated IDs directly. A geometric mean across prompt summaries is used for headline speedup. B570 timing remains diagnostic regardless of its estimator.

The B570 finalizer builds its prompt vector in the insertion order of the archived pair records. Sorting by task ID changes the finite fixed-seed bootstrap endpoints even though it leaves the underlying estimator unchanged. Exact replay retains that original order. The M4 HumanEval lower endpoint uses a different order statistic from M4 24Q and RTX; those frozen conventions are retained. None of these prompt intervals captures the full distribution of possible thermal, session or runtime states.

C ENERGY BOUNDARIES AND ALLOWED CLAIMS

Ratios above one favor speculation within the named boundary. For M4 fixed-$k = 2$, the package ratio is 1.013 [0.990, 1.034], so a package-energy improvement is not established. The registered $p_{\min} = 0.9$ secondary condition has a package-efficiency ratio 1.041 [1.018, 1.065]. This is not a replacement for the primary condition and does not establish a material additional speed gain. B570 serialized board efficiency is 0.302 [0.285, 0.320]; correctness failure restricts it to diagnosis. No between-platform energy ranking is valid.

D ROUND ACCOUNTING AND STATE SEMANTICS

The 358H target keeps its low-level context across rounds. Candidate evaluation, invalid-suffix rollback when needed, and correction or bonus decoding belong to the complete target-side timing region. Every OpenVINO proposal call receives the entire committed prefix. Its synchronous timing combines reconstruction and proposal; neither term can be isolated from this experiment without changing the measured execution.

The CPU path records 11,520 committed output tokens over 4,677 rounds; the NPU path records 11,520 over 4,686 rounds. Their directly measured C values are 2.4631 and 2.4584. Using the arithmetic baseline time per output token and arithmetic component totals yields CPU $(D, R, X) = (0.4304, 2.9767, 0.00124)$ and NPU $(4.1251, 2.9435, 0.00130)$. The resulting ratios 0.7227 and 0.3477 match ratios of total times, not the prompt-GM headline estimator.

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Table A1. Frozen statistical policies. GM = geometric mean. Indices below are zero-based order statistics of sorted bootstrap draws; NumPy quantiles use the original analyzer's convention.

Campaign	Within prompt	Resamples	Seed	CI endpoints / generator
M4 24Q	GM of 3 ratios	10,000	20260907	[249,9749]; Python Random
M4 HumanEval	GM of 3 ratios	100,000	20260910	[2500,97499]; Python Random
RTX HumanEval	GM of 3 ratios	10,000	20260910	[249,9749]; Python Random
B570 fallback	GM of 3 ratios	10,000	20260910	[250,9750]; Python Random
358H Phase 3	Median of 3 ratios	10,000	20260919	Quantiles .025/.975; NumPy default_rng

Table A2. Sensor boundaries and allowed interpretations. No cells are combined into whole-system energy.

Campaign	Measurement boundary	Allowed statement
M4 HumanEval	Silemetry CPU + GPU + ANE package aggregate; separate DRAM excluded	Fixed-$k = 2$ package-energy interval includes parity; registered threshold variant secondary
RTX 5070 Ti Laptop	NVIDIA board / CPU package are distinct domains; follow-up process window includes loading	No formal serialized HumanEval energy comparison
B570 fallback	Arc adapter board only; no CPU package or wall power	Diagnostic board efficiency only; no workload-global text-path agreement
358H Phase 3	No qualified end-to-end energy boundary	No end-to-end CPU claim; NPU energy not admissible

Historical M4/RTX accepted-draft counters exclude correction or bonus output and have output-boundary effects. They are not substituted for the directly recorded 358H committed-progress metric. Likewise, target-forward PP2/PP1 ratios are never presented as complete realized target-round cost.

E Deviations and Provenance Boundaries

The M4 corrected synthetic ceiling was preregistered but not executed. Earlier synthetic acceptance diagnostics are not a substitute. The representative and separately preregistered code workloads stay distinct. Early MLX precision/correctness failures are qualification history, not failures of the final Metal Q8_0 path.

The B570 formal v2 correctness failure triggered its stop rule. A later six-prompt serialized probe passed 18/18 text-path comparisons, but full-workload serialization yielded only 60/90 under the same comparator. Earlier remote-interaction-contaminated verification observations are excluded from the plotted final serialized cells. A legacy "exact" field label in historical artifacts does not establish workload-global equality.

Windows follow-up derived tables and numerical captures support the manuscript, but historical raw per-run provenance is weaker. The CUDA score capture came from a separate same-commit instrumented build and does not replace formal measurements. The 358H source reconstruction is strongly supported by retained evidence but lacks proof of exact historical upstream identity (the internal `STRONGLY_SUPPORTED` label). Its completed Phase-3 records supersede the blocked-preflight empty scaffold. These qualifications describe the limits of the retained evidence.

The historical RTX HumanEval/106 CUDA reproducer selects token 2790 under sequential/serialized execution and 2629 under batching in three of three runs at the same location. A separate same-commit diagnostic build captures the score-margin change from $+0.080948$ to -0.126457. This supports stable batch-shape-dependent greedy argmax divergence in that historical configuration; it does not establish a specific kernel, universal CUDA defect, or a shared RTX/B570 root cause.

F Historical Proxy and Separate Qualification

The historical main Figure 3 is relocated here with its PDF asset, inclusion width and caption unchanged (Figure 1). This figure is background: its panels do not constitute a

matched qualification/cost study. For context n and execution path e, the target-forward-only proxy is

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}.$$

It omits drafting and may omit correction/bonus and state work; it is not complete target-side round cost. Original RTX and B570 direct-ID campaigns each use 30 prompts $\times$ three, no timing and a 90/90 gate. RTX batching is 75/90, serialization 90/90; B570 fast is 60/90, serialization 78/90. These are different campaigns from historical timings, B570 H1 and RTX V4; no retrospective qualification or pooled tradeoff follows.

Table A3 retains all historical context cells behind Panel A of Figure 1. Values are target-forward ratios, not complete target-side round cost or integrated speedup. Context ranges do not represent uncertainty intervals. RTX denotes RTX 5070 Ti Laptop throughout this appendix.

G Evidence Inventory

Table A4 collects only the original case inventory and evidence classes; follow-up cohorts and endpoints are documented in Sections K–N. It is not an inventory of all newer prediction studies. These are case descriptions, not a hardware ranking, population-wide failure rate, or exhaustive taxonomy. M4's separate-session studies establish opposite observed outcomes; neither diagnosed mechanism is assigned to that contrast.

H Runtime Study Reproducibility

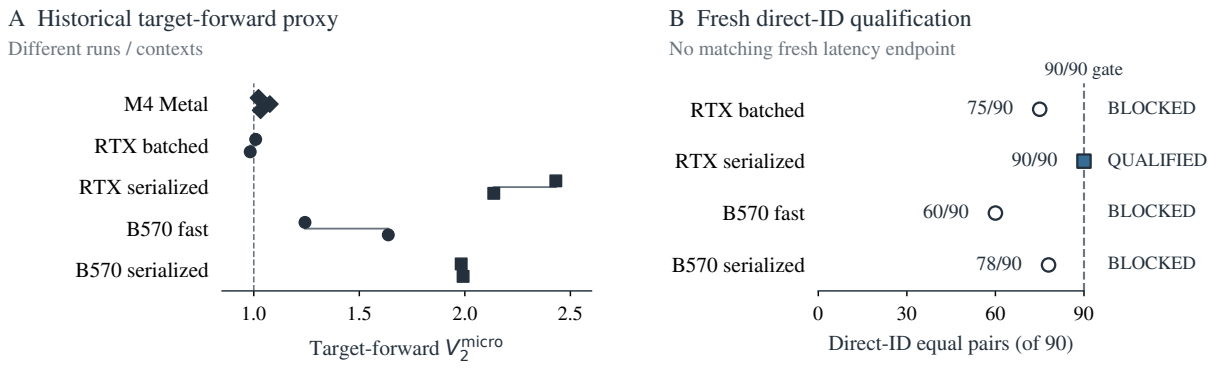
The M4 runtime estimator first takes each condition's median of 3 repetitions, then forms a prompt latency ratio and its geometric mean across 60 prompts. The offline oracle uses the smaller target-only or Always-Spec median for each prompt. Bootstrap intervals use 10,000 draws with Python Random seed 20260924, resampling prompts in the frozen held-out order. Percentile endpoints linearly interpolate the sorted bootstrap draws at quantiles .025 and .975. A single random stream computes Target/Guard, Target/Always-Spec, and Guard/oracle intervals in that order. This preserves the frozen finite-bootstrap calculation; it does not replace the hardware estimators in Table A1. Intervals condition on the selected policy and observed prompt medians; they do not incorporate recalibration or between-session variation. The oracle selects between two measured fixed conditions, not all possible switching policies, so it is not a universal latency lower bound.

Calibration, qualification, and held-out IDs are disjoint. Selection ranks tasks with no recorded prior M4 exposure in the frozen inventory by a seeded hash within each of HumanEval and Spec-Bench; each split is balanced across these two families. The registered grid has nine candidates: first decision round $E \in \{2, 4, 8\}$ and margin $m \in \{0, 0.05, 0.10\}$. Calibration selects the exact candidate with the lowest geometric-mean latency ratio to the offline oracle. Exact ties break by mean measured guard-compute overhead, smaller exploration count, larger margin, and lexicographic candidate ID. The frozen winner is `E8_M000`; held-out results never select or retune it.

The portable M4 archive contains original observations with generated-token IDs, complete-attempt manifests, qualification and calibration records, frozen source/build information, and analysis. An independent offline verifier reproduces policy selection, qualification and formal direct-ID equality, prompt medians, ratios, intervals, and fallback fraction without executing the models. No runtime energy result is claimed.

The reviewer-requested Guard/Always-Spec contrast is post-study paired secondary analysis, not the registered primary endpoint. Aggregate results were already known; the committed analysis plan is a procedural timestamp, not preregistration or blinding. For each frozen held-out prompt i, form $r_i = \text{median}(T_{\text{Guard},i}) / \text{median}(T_{\text{Always},i})$ over its 3 repetitions, then compute $\exp(60^{-1} \sum_i \log r_i)$. Values above one mean Guard is slower. A fresh standalone Python Random stream with seed 20260924 resamples 60 paired prompts per draw in frozen held-out order, using 10,000 draws and the same linear percentile interpolation; it does not continue the primary stream. The ratio is 1.0271 [1.0130, 1.0436], with the interval above parity. Descriptive prompt counts are 24 Guard-faster, 36 Always-Spec-faster, and 0 exact ties. Both implementations reconstruct raw medians from the same hash-verified archive and match the primary medians exactly; an independent calculation reproduces both finite-bootstrap endpoints. No new measurement was made. The comparison and its analysis choices were fixed for this secondary replay after aggregate results were already known; they were not blinded or prespecified in the original experiment. No choices were changed during that replay to improve the result. The v0.8.6 sensitivity below is a separate, explicitly post-study choice. The interval excludes session and recalibration variation; source hashes and all prompt ratios remain in the private evidence ledger.

A metadata-recording defect requires explicit provenance recovery. Loop-variable shadowing caused the frozen helper to return the build-manifest path rather than the calibration-freeze path. Qualification and formal-start records therefore contain the build-manifest digest in the intended freeze-hash field. A versioned recovery analyzer verified the actual freeze against the clean formal-start commit, qualification, frozen source/model/calibration/policy hashes, and complete attempts before applying the original formulas. The archive retains the original error markers and pipeline



A: path shapes only; all 11 measured cells. B: open = BLOCKED; filled square = QUALIFIED.

Figure 1. **Historical target-forward proxy and fresh direct-ID qualification are separate evidence.** A: historical V_2^{micro} cells from different runs and contexts; shapes encode M4 reference (diamond), fast/batched (circle), and serialized (square) paths, with no qualification encoding. Segments span measured contexts, not uncertainty. The proxy is not complete R_{target} or end-to-end speedup. B: fresh HumanEval30 direct-ID sequence equality, with no matching fresh latency endpoint; only RTX serialization reaches the frozen 90/90 gate. Open circles mean BLOCKED; the filled square means QUALIFIED. The panels are not a matched cost–qualification experiment. Fresh direct-ID qualification does not retrospectively authorize historical timing. Results are path/runtime scoped; contexts and references differ.

`STOPPED` state, meaning automated finalization stopped at the metadata exception. The scientific policy itself was not retuned. Its actual calibration-freeze digest is `3c179a35f046b57edb4ddc8772a1aba609b496878efc2b7849a66947c8b7830b`; the misrecorded build-manifest digest is `21592be38c0c477f63cea819b96b5b12cfea4529c131f6914b761a06ae5e518b`. These checks validate retained links and recorded ordering; they do not supply trusted external time. This is a metadata-recording defect, not a measurement mutation.

The 358H diagnostic is pinned separately from the hardware campaign. Its compact evidence records reproducible target-only same-prefix divergence and proposal playback without live NPU inference, but does not establish a specific rollback-boundary bug. It explicitly records that no formal runtime performance data exist and that failed qualification prevented formal runtime performance evaluation. Internal evidence records retain the exact source commit and file hashes; the local raw diagnostic traces are not included in that compact record. No 358H runtime speedup is inferred from these diagnostics.

I Prospective Direct-ID Qualification

Table A5 reports fresh direct-ID qualification separately from the historical text-path lookup and timing inventory. Both studies freeze the same HumanEval30 IDs, Qwen3-8B and Qwen3-0.6B Q8_0 models, greedy $k = 2$, three repetitions, cap 128, and a 90/90 complete direct-ID sequence-equality rule. Prompt IDs and uncommitted proposals are excluded; an earlier generated terminal EOG is included. No mismatch is discarded and no subset qualifies. Historical path-control outcomes informed the replications; prospective freezing does not imply blinded discovery. Each study completes 270 executions; independent offline implementations reproduce all 180 comparisons, including first divergences. Operational elapsed fields do not establish formal performance.

The RTX original embedded-server CLI exposes native committed-token debug events without a source patch. It uses context 4096, reference target micro-batch one, candidate micro-batches two/one, driver 616.92, and full CUDA0 offload. The historical RTX driver was 616.56; this is not an exact replay of historical host state. The incomplete RTX v1 transport attempt is preserved and supplies no v2 formal pair; completed v2 uses lossless byte transport, a new prospective freeze, and a complete fresh schedule.

B570 uses source instrumentation to record committed IDs in sidecars, context 1024, and a separately frozen completion reference. Its endpoint selects the first 128 committed IDs or the inclusive earlier EOG; raw CLI overshoot is retained. Vulkan0 uses driver 32.0.101.9033 with full target 37/37 and draft 29/29 layer offload. The initial Task Scheduler launch failed during CP936/UTF-8 environment-metadata decoding before inference, with zero formal executions. The first data-producing attempt directly ran the unchanged frozen runner: no semantic, code, binary, model, driver, or parameter change occurred. This deployment deviation is preserved in provenance.

Table A3. Historical target-forward lookup. Each row is a measured context cell; ratios are shown to six decimal places from the frozen source. Status refers to historical text-path agreement, not fresh direct-ID qualification. M4 and B570 contexts are prefix lengths; RTX uses reported context depth. The stacks and contexts are not matched. Rounded ranges are M4 Metal 1.02–1.08, RTX batched 0.98–1.01, RTX serialized 2.14–2.43, B570 fast 1.24–1.64, and B570 serialized 1.98–1.99; they span contexts, not confidence intervals.

Path	Context (tokens)	V_2^{micro}	Historical text-path status
M4 Metal	512	1.022182	Text-path agreement; shared path
M4 Metal	2048	1.076126	Text-path agreement; shared path
M4 Metal	8192	1.032264	Text-path agreement; shared path
RTX batched	0	1.009030	Text-path blocked; diagnostic
RTX batched	256	0.983148	Text-path blocked; diagnostic
RTX serialized	0	2.430996	Text-path secondary control
RTX serialized	256	2.137011	Text-path secondary control
B570 fast	512	1.242576	45/90 text-path; diagnostic, inadmissible
B570 fast	2048	1.637295	45/90 text-path; diagnostic, inadmissible
B570 serialized	512	1.982668	60/90 text-path; diagnostic, inadmissible
B570 serialized	2048	1.991707	60/90 text-path; diagnostic, inadmissible

The runtime/front-end, reference, and context differences preclude a causal hardware comparison. Neither study estimates population failure rates or functional code correctness. Repeat stability holds for every prompt in each condition, but stable disagreement remains disagreement. Historical B570 fast 45/90 and serialized 60/90 text-path counts and historical RTX controls retain their original comparators. New qualification does not retrospectively authorize old timing as strict-ID performance. Portable validators hash included evidence and check recorded identities of omitted external weights/binaries; they do not rehash omitted assets locally.

J B570 Target-Partition Prediction

The B570 Vulkan follow-up uses Qwen3-8B/0.6B Q8_0, greedy $k = 2$, context 1024 and 128 committed IDs or earlier inclusive EOG. Its target-only harness matches the UB128 reference's input IDs, sampler and APIs; H1 changes only target n_{ubatch} to one. The harness reproduces 90/90 reference sequences before comparison with live serialized speculation.

Selection and freeze. The audit covers 8,706 retained file records and archive members; any B570 association is conservatively excluded, even if execution is uncertain. This excludes 31 of 164 tasks. The 24 reserved IDs have the smallest SHA-256 digest of a fixed study prefix plus each eligible task ID; outcomes play no role. Discovery uses exposed HumanEval/89, 101, 104, 162, 3, 10, 68, 70, 19, 20, 28, 32. The frozen tree selects H1 at 36/36; only failure permits H2 fixed-proposal replay (also 36/36), and failure of both stops validation. H1 passes; H2 is not built. Predictor, endpoint, reserved IDs, binaries and criterion are committed before reserved inference.

Prospective validation. The 24 reserved IDs are HumanEval/58, 156, 134, 42, 122, 114, 93, 115, 151, 63, 125, 133, 43, 139, 77, 35, 27, 34, 145, 82, 154, 78, 96, and 138. Target reference and target-only predictor run three times each (144 executions) with within-condition repeat stability. All 72 predicted complete ID arrays and their digest are committed before the first of 72 live serialized outcomes. Primary prospective trajectory validation requires 72/72 predictor-to-live complete-ID agreement; no partial threshold or confidence interval is substituted. That agreement is 72/72. Twenty prompts are predicted reference-equal; four are predicted divergent, and all 24 classifications match. Among the 12 divergent repetition pairs, the first-divergence index matches 12/12. Those four prompts and zero-based indices (reference token → serialized token) are HumanEval/134: 88 (914→1946); /42: 73 (432→1817); /145: 103 (1184→686); and /138: 93 (14374→9064). Reference-to-live direct-ID qualification is 60/72 and BLOCKED on this reserved set.

An independent validator passes 5,681 input, output, hash, endpoint and chronology checks; extracted archive validation passes. No prediction edits, selective reruns, parameter changes, performance or energy experiments occur after the relevant freezes. No-known-exposure is limited to audited records. The result establishes tested predictive path sufficiency, not a unique kernel, universal rule, CUDA/B570 shared cause or production certificate.

Reviewer-visible commitment record. The comparison uses complete generated-token-ID arrays; prompt IDs are excluded. At most 128 committed generated IDs are compared, including an earlier terminal end-of-generation (EOG) ID when it ends generation. No mismatch is discarded. Equality requires the complete sequences, including lengths, to

Table A4. Evidence inventory with row-level output endpoints. P: primary; S: secondary; M: mechanism; D: diagnostic. Historical M4/RTX/B570 rows report text-path agreement; 358H reports direct-ID sequence equality. Both B570 paths remain inadmissible diagnostics. The RTX serialized controls each have 90/90 historical text-path agreement. The 358H P/M entries separate primary outcomes from mechanism-supported decomposition. None establishes functional code correctness; annotations are descriptive and non-exhaustive.

Case	Evidence endpoint / gate	Outcome	Class	Case annotation
M4 representative 24Q	Text-path 72/72	Negative	P	Workload-conditioned
M4 HumanEval30	Text-path 90/90	Profitable	P	Workload-conditioned
RTX batched	Text-path; BLOCKED	Not admissible	M	Verification-constrained
RTX serialized controls	Text-path 90/90 each	Negative	S	Verification-constrained
B570 fast	Text-path 45/90	Not admissible	D	Correctness-blocked
B570 serialized	Text-path 60/90	Not admissible	D	Correctness-blocked
358H CPU	Direct-ID 90/90	Negative	P/M	Target-side dominated 87.34% target side
358H NPU	Direct-ID 90/90	Negative	P/M	State-interface bound 58.35% replay + proposal

Table A5. Fresh direct-ID qualification studies. Each path must achieve 90/90 complete generated-ID sequence equality; a wholly equal prompt matches in all three repetitions. QUALIFIED/BLOCKED is the frozen gate result. These studies have no matching fresh latency endpoint.

Platform	Path	Direct-ID pairs	Equal prompts	Gate	Evidence endpoint
RTX 5070 Ti Laptop	Batched	75/90	25/30	BLOCKED	Direct-ID sequences
RTX 5070 Ti Laptop	Serialized	90/90	30/30	QUALIFIED	Direct-ID sequences
B570	Fast	60/90	20/30	BLOCKED	Direct-ID sequences
B570	Serialized	78/90	26/30	BLOCKED	Direct-ID sequences

match; first-divergence indices are zero-based.

Content identity. The 72 frozen complete prediction arrays are bound to the following prediction-file content digest (SHA-256; two consecutive 32-character lines):

```
0201ccd1f67492510d34be567aa83f1c
38d3bd1f162a970c05d3cea9d3957826
```

The digest identifies the frozen contents; the temporal ordering below comes from the recorded study chronology rather than from the digest alone.

Recorded event	Time relative to H1 freeze
H1 freeze	T_0
72 prediction arrays finalized	$T_0 + 1722.38\,\mathrm{s}$
First live reserved outcome	$T_0 + 1813.75\,\mathrm{s}$

The recorded study chronology fixes the prediction file before the first live serialized reserved outcome, with a 91.37 s interval between prediction completion and live start. Identifying repository metadata are omitted for double-blind review.

Informative reference-divergent prompts. These cases are reference-divergent but predictor/live-identical.

Table A6. Reference trajectory agreement versus frozen UB1 predictor agreement on the same 24 reserved prompts × 3 repetitions. Predictor/live complete-sequence equality is the primary prospective path-prediction endpoint; reference/live equality is an already-observed comparison. The 60/72 reference result remains BLOCKED under its qualification contract.

Comparator to live serialized output	Complete equality
Frozen UB128/reference trajectory	60/72
Frozen UB1 target-only predictor	72/72

Task	First index	Ref → Pred/Live ID	Pred=Live
HE/134	88	914 → 1946	3/3
HE/42	73	432 → 1817	3/3
HE/145	103	1184 → 686	3/3
HE/138	93	14374 → 9064	3/3

Across these 4 prompts, the predicted and observed first-divergence positions agree in 12/12 repetition pairs.

The separate anonymous sequence supplement contains

Table A7. B570 reserved commitment ledger. All 24 prompts have three repeat-stable 128-ID outputs. Pred=Live and Ref=Live count complete generated-ID equality across three runs. Each displayed digest is a 12-hex prefix of the full SHA-256 shared by predictor and live output; complete arrays and hashes are in the anonymous sequence supplement. The first reference-to-live divergence index is zero-based; – means reference equality.

Task	Live len	Pred=Live	Ref=Live	Pred/Live digest	Ref → Live first divergence
HE/58	128	3/3	3/3	cc12b5e1b600	–
HE/156	128	3/3	3/3	b8060d33b72c	–
HE/134	128	3/3	0/3	fc12c8453de5	88: 914 → 1946
HE/42	128	3/3	0/3	c737973a39a6	73: 432 → 1817
HE/122	128	3/3	3/3	ed3172303704	–
HE/114	128	3/3	3/3	16037f527b2f	–
HE/93	128	3/3	3/3	4b69a84d3675	–
HE/115	128	3/3	3/3	43fbb9bd3da9	–
HE/151	128	3/3	3/3	f371e4f2b94e	–
HE/63	128	3/3	3/3	b3141572ec95	–
HE/125	128	3/3	3/3	689af3f02f2d	–
HE/133	128	3/3	3/3	c1ed3bf83bde	–
HE/43	128	3/3	3/3	2e3776c7ff02	–
HE/139	128	3/3	3/3	23a77663974d	–
HE/77	128	3/3	3/3	dea569b38784	–
HE/35	128	3/3	3/3	93498855fe64	–
HE/27	128	3/3	3/3	f0f161a1b9e3	–
HE/34	128	3/3	3/3	813cb3180ab5	–
HE/145	128	3/3	0/3	31246ffe07b9	103: 1184 → 686
HE/82	128	3/3	3/3	a356c9235624	–
HE/154	128	3/3	3/3	0743671d064c	–
HE/78	128	3/3	3/3	a685f47dbab9	–
HE/96	128	3/3	3/3	3824c19fc287	–
HE/138	128	3/3	0/3	9e54992a7160	93: 14374 → 9064

complete reference, predictor, and live generated-ID arrays for the 72 reserved runs, with a deterministic equality and divergence checker. It contains no identifying machine or repository metadata. Sanitizing the export changes its bytes, so it cannot reproduce the original prediction-file digest or independently prove the reported temporal freeze; the PDF's digest and chronology have those distinct stated roles.

K Follow-up Campaign Identities

The follow-ups retain the Qwen3-8B/0.6B Q8_0 pair and upstream llama.cpp b10839, but not one common frontend or reference. Table A9 identifies the new campaigns. Original RTX direct-ID qualification used Windows build 26200; new RTX H1, latency and H2 use 26300. Their driver version is 616.92, versus historical timing's 616.56; version equality does not authenticate original driver bytes. Recorded byte identities cover models, frontend, DLLs, compiler and observed driver modules on the experimental host; omitted assets cannot be rehashed by an offline reader.

Separate selection and exposure. Original B570 H1's 24 reserved tasks and 133-task eligible pool remain as documented above. The live fork study excludes old direct-ID, old reserved and other recorded exposure before deterministic selection using its fixed study salt. The RTX H1 arm excludes 103 exposed or ambiguous tasks from 164, leaving 61; its reserved list is never executed after discovery failure. V2 re-audits accessible scientific files, archives and intervening study records: 114 tasks are excluded, leaving 50. Its selection uses the smallest SHA-256 values of a fixed V2 salt plus full task ID, without outcome selection. The actual 24-task V2 order is /160, /45, /149, /17, /6, /72, /33, /121, /130, /12, /4, /55, /69, /14, /61, /135, /100, /117, /22, /7, /23, /95, /88, /37. These tasks must not be pooled with original B570, the new fork set, the unexecuted H1 reserved list or latency's HumanEval30. Offline selection checks reconstruct retained inventories, not a new scan of unavailable remote directories.

L B570 Same-State Fork Endpoints

The frozen checkpoint is the first native verification batch beginning at at least 64 committed IDs. It is one sampled block per eligible execution, not every generation position. All 72 reserved executions have eligible forks. The live draft's two proposals are recorded before either target decode. Source KV is serialized, restored into the alternate context, serialized again and compared with source-after bytes; consumed length, prefix, positions and physical groups are checked. Native UB1 produces three one-output physical groups, whereas alternate UB128 produces one three-output

Table A8. Historical case-study settings; follow-up campaign configurations are listed separately in the appendix. These are not a matched hardware comparison. All final target models are Qwen3-8B Q8_0. Greedy execution uses campaign-matched target-only references; historical M4/RTX/B570 use re-tokenized text, while 358H uses direct IDs.

System	Runtime / target backend	Draft model / placement	Evidence scope
MacBook Air M4	llama.cpp b10809 / Metal	Qwen3-0.6B Q8_0 / Metal	Representative 24Q and separate HumanEval30
Core Ultra 9 275HX + RTX 5070 Ti Laptop	llama.cpp b10839 / CUDA	Qwen3-0.6B Q8_0 / CPU or same GPU	Blocked fast path; text-path-agreeing controls
Core i5-13400 + Arc B570	llama.cpp b10839 / Vulkan0	Qwen3-0.6B Q8_0 / B570	Formal gate failure; later diagnostics
Core Ultra X7 358H + Arc B390	llama.cpp b10839 / Vulkan0; OpenVINO draft	Qwen3-0.6B symmetric INT4 / CPU or NPU	Closed Phase 3; direct stateful decomposition

Table A9. Campaign-specific follow-up configuration. RTX always denotes RTX 5070 Ti Laptop. Greedy target sampling uses seed 123, FA off, fresh contexts, and an inclusive EOG/128 committed-ID endpoint. All layers are offloaded: target 37/37 and draft 29/29. The original studies' references remain unchanged. V4 is documented separately in Section Q.

Campaign	Frontend / execution	Reference or paired contrast
B570 live fork	Instrumented native speculative example; Vulkan0; context 1024, batch 512	Native UB1 versus state-restored UB128; identical fixed proposals at sampled block; UB128 continuation persists
RTX H1 discovery	Original CLI/CUDA DLLs; context 4096	Target-only UB2 versus batched native live; 24/36 fails selection, reserved 0
RTX H2 V2	Separate P/R workers and original native CLI; shared numerical DLLs; context 4096, batch 2048	H2 and R_ONLY target UB2; original target-only reference UB1; native live target/draft UB2
Earlier RTX two-path cost (30 prompts)	New Clang 21.1.0 timer frontend/core, original hashed CUDA backend DLLs; context 4096	Same instrumented executable for target-only UB1 and serialized UB1; new 90/90 direct-ID gate

group at this sampled block.

Sampler identity uses the pinned recursive `common_sampler_clone`, copied chain/seed/accepted-token history and the same-partition R0 control. Six exposed engineering tasks generate 24 fresh R0 executions; original/instrumented primary IDs and proposals agree, and UB1-to-UB1 restored continuation and scores agree. Opaque sampler serialization is unavailable, so these are source-audited clone and empirical controls, not a byte proof for omitted external sampler binaries.

The primary comparison follows every reachable decision at a still-common proposal-conditioned prefix in the sampled block until rejection, bonus or endpoint: 195 paired events. All three raw score rows are retained even when proposals are rejected. Score equality, raw argmax equality and actual accepted/corrected/bonus IDs are different tests. The exported paired records preserve them separately.

After the native continuation ends, the implementation restores the saved draft KV and cloned samplers, replaces target context with the already-verified alternate UB128 context, restores counters and committed prefix, then continues the native loop. Subsequent target blocks remain UB128, and subsequent draft proposals follow alternate accepted history. This continuation policy changes more than a single numerical block over the rest of generation. The 18/72 trajectory differences across six prompts occur after unchanged sampled immediate decisions; they cannot be attributed to an immediate split at that block or claimed as reference restoration. At the tested states, the policy changes scores and later trajectories, with no detected immediate de-

Table A10. B570 reserved fork results: 24 new prompts × three. Events and repetitions are correlated. Primary immediate decisions remain unchanged; later output differences are a separate secondary endpoint.

Endpoint difference	Runs	Prompts
Raw score records	72	24
Raw top-1 IDs	0	0
Immediate decisions	0	0
Full continuation IDs	18	6

cision effect. These results identify no unique operation, do not rule out effects elsewhere in generation and support no RTX or performance inference.

M RTX Separate Proposal/Replay Orchestration

The earlier failed arm. H1 target-only UB2 matches 24/36 newly executed exposed batched trajectories. /64, /106, /119 and /128 each fail all three repetitions. The 36/36 gate is unmet, no predictor is selected, no hypothesis or prediction freeze is reached, and reserved runs are zero. H2 was not implemented in that frozen arm. V2 is a new study, not a retroactive execution of the earlier decision tree.

Program independence and shared computation. P loads only draft weights; R loads only target weights. H2 orchestrates these processes, first recording each proposal request and then sending it to R. P uses H2's already-produced feedback; it is not a feedback-free draft tape. R computes pending tokens, acceptance, correction, bonus and KV suffix removal. Fresh R_ONLY reads one sealed tape request at a time and checks its own computed prefix and pending token. Future prediction/reference/live IDs are excluded from its computation inputs. The original native CLI produces its own proposals and has no tape import. Model-loader, compile-time role, allowed RPC fields and process/load records establish application/API isolation, not adversarial OS-wide file access. Original model loaders, samplers, llama.cpp/GGML/CUDA kernels are shared, so there is no independent numerical implementation.

Freeze chain and denominators. F0 seals source, effective configuration, input/exposure/selection and qualification before discovery. Discovery completes 108 executions (12 tasks $\times$ three repetitions $\times$ H2, R_ONLY and native live), with both H2/live and H2/R_ONLY 36/36. F1 seals the selected program before 216 reserved pre-live executions (reference/H2/R_ONLY). F2 seals the 72 prediction rows, full tapes and pre-live package; remote acknowledgement and retrieval of the exact sealed bytes precede the first of 72 reserved native live runs. Thus formal four-condition execution totals 288, with 72 comparisons per named endpoint. Git blob/tree membership certifies preserved bytes, while server and host records provide the reported order; neither is a trusted per-process external clock. Sanitized anonymous ordinals cannot independently prove that original absolute chronology.

Actual endpoint distribution. H2, R_ONLY and native live each contain 63 earlier-EOG runs on 21 prompts and nine cap runs on three prompts. The UB1 reference has 66 earlier-EOG runs on 22 prompts and six cap runs on two prompts. Each includes the first EOG ID and excludes prompt and uncommitted proposals. This differs from original B570 H1, where every reserved array reaches 128. Repetition stability holds within all four conditions; it is not reference equality. The reference gate remains blocked. There is no target-only UB2 competitor on these same V2 reserved prompts, so these matches do not establish proposal/history necessity.

Table A11. RTX V2 reserved equality. Every row uses 24 distinct prompts $\times$ three repetitions. Prediction equality and reference qualification have different comparators.

Comparison	Equal / 72	Prompts / 24
H2 / native live	72	24
R_ONLY / native live	72	24
H2 / R_ONLY	72	24
UB1 / native live	60	20

Table A12. RTX V2 reference-divergent cases, stable in all three repetitions. Indices are zero-based generated-ID positions; H2 and R_ONLY predict the native live value. Reference is this study's UB1 path.

HumanEval	Index	Reference ID	Live ID
/17	4	1406	271
/33	8	14937	4210
/14	4	1406	271
/22	0	1499	750

N Fresh RTX Serialized Cost Estimator and Timer

Thirty distinct prompts from the latency arm's frozen original HumanEval30 input list each receive three executions of target-only and serialized: 180 new runs, 90 direct-ID pairs. Conditions are randomly ordered within prompt/repetition with seed 2026093001; bootstrap seed is 2026093002.

For condition medians m_{ic} on prompt i, compute

$$r_i = m_{i,\text{target}}/m_{i,\text{serialized}},$$
$$G = \exp\left(\frac{1}{30}\sum_{i=1}^{30}\log r_i\right).$$

The 10,000 bootstrap draws resample 30 prompts in original numeric task order using Python Random, with linearly interpolated sorted percentiles at .025/.975. This is neither median of paired ratios nor ratio of 180 total times; repeats are not bootstrap units. Independent full-precision replay $G = 0.6458157742698285$ (the sealed original report stores 0.6458157742698286, one ulp higher), CI [0.629802205379344, 0.6598843928244866], with 0/30 ratios above one. The independent floating-point replay

agrees within 10^{-14}; full-precision canonical and independently recomputed results remain separately recorded. The rounded manuscript value is 0.646 [0.630, 0.660].

Both conditions use the same freshly built timer executable and pass their own 90/90 direct-ID gate. The native QPC interval begins after synchronized target/draft initial prefill and before per-request sampler creation/reset/seed and generation-state work. Draft inference, target validation, acceptance/correction/bonus, KV state, synchronization, proposal/scheduling and relevant final response/debug-log work through slot release are charged. Loading, initial prefill and teardown are excluded. Persistent engine allocation and one-time draft sampler precede prefill; request sampler work is inside the timer. Endpoint records print after measured end. A second chrono interval cross-checks the native endpoints, but both APIs share Windows' clock oscillator.

The frozen empty-hook probe checks clock-call overhead at its prescribed threshold; it does not show that total committed-ID/debug logging costs less than 1%. This is instrumented CLI cost, not deployment throughput. Thermal entry and randomized adjacent scheduling reduce drift without locking all clock/session variation. No energy result is estimated. The earlier RTX timing ratio 0.650, Figure 3 proxy cells and qualification-only gates are unchanged. The study has no qualified batched timing endpoint, so it does not measure a full batched-to-serialized speed tradeoff.

N.1 Proposal inputs and local decisions

B570 H1 computes a target-only UB1 sequence with no proposals. RTX H2 instead constructs a feedback-dependent tape using draft-only P and target-only R; both share llama.cpp/GGML/CUDA. The sealed implementation uses this flow:

```
P(past feedback) -> proposals
R(own state, proposals) ->
decisions
seal tape; fresh R_ONLY -> own
decisions
```

R initialization accepts only the formatted prompt; its verification allowlist is `op, round, position, pending, proposal_ids`. Round, position and pending must match R's locally maintained round, KV position and last sampled committed ID. Tape task/repetition, prefix count/hash and request dependencies are checked against the replay's own accumulated prefix. They are assertions of past state, not supplied future answers. Each tape row is consumed only after the previous verification completes; truncation, hash-chain mismatch, state mismatch, extra fields or a suffix after the endpoint aborts execution. R samples target logits locally, accepts matching proposals, emits correction at rejection or bonus after full acceptance, and removes the invalid suffix. P updates from its own verified batch and R's already-produced retain count. Live CLI generates its own proposals and imports no tape.

For reserved HumanEval/17 repetition 1, round 0 requests proposals [19496,1159] at position 195 with locally pending ID 1499. R samples [19496,1159,1759], accepts two proposals and commits bonus ID 1759 at generated index 3. Round 6 requests [4210,2505] at position 213, pending ID 262 and prefix length 19; R samples ID 8388, rejects the first proposal, retains one slot and commits ID 8388 at index 19. Both slices equal the final H2/R_ONLY/live arrays. Source allowlists and sampler calls support the input separation; array equality alone does not prove non-leakage or independent numerical implementation.

The retained engineering capsule contains five modified-file witnesses (changed committed ID, deleted event, task mismatch, tape-round/hash mismatch, future-answer field) and a premature-live ordering witness, plus constructed acceptance/bonus and proposal-width fixtures. These are existing records, not new model tests. Offline rejection checks strengthen file/protocol review; they do not establish adversarial OS isolation. Process/model-loader records limit independence to application/API orchestration.

O Reviewer Evidence and Offline Recalculation

The H2 capsule's SHA256 is the full filename digest in Table A13, and its length is 60,152,666 bytes. `INDEX.json` gives separate-capsule identity and study-specific source/policy → predictions/tapes → acknowledgement → first-live → analysis roles. Original H1 has separate hypothesis and 72-array freezes; B570 forks freeze sampled policy before reserved execution; RTX H1 stops at discovery; RTX V2 F0/F1/F2 seal configuration, selected program, then predictions/tapes. RTX cost freezes its new timer and protocol before qualification/formal measurement; M4 policy recovery is documented in Section H. New descriptive analyses and exports follow known outcomes and claim no pre-live freeze.

Original B570 arrays are compared element by element with sealed originals; the exact earlier sequence ZIP is unavailable and is not replaced. The compact index adds post-study analysis inputs and a standard-library calculator while retaining existing arrays. Original host paths, identities, Git authors and absolute timestamps stay outside reviewer exports; internal reviews are excluded. Content digests may permit linkage and do not guarantee unlinkability. Retained records support reported original ordering; sanitized ordinal mapping cannot authenticate it independently. Omitted weights/runtime/compiler/driver bytes and full external as-

Table A13. Evidence capabilities and exact reviewer entries. The source ZIP reconstructs documents; evidence packages recompute retained observations without models. Derived hashes/ordinals cannot certify original absolute chronology or omitted binaries.

Object	Exact package / entry	Capability and limit
Document sources	`overleaf-system-revision-v0.8.9.zip`; `latexmk -pdf main.tex`, then `appendix.tex`	TeX dependency/hash audit and fresh build; no science execution
Compact studies (including RTX V4)	`anonymous-evidence-index-v0.8.8.zip`; `python3 -B verify.py`	Direct CPU recomputation of retained arrays, timer intervals and post-study summaries; not raw state bytes or absolute time authentication
RTX full anonymous capsule	`anonymous-a5919b59342ecc68bd5f57d3c2b192edf62933b2645924381c3765dd50beb369.tar.gz`; extract and run `frozen_source/scripts/platform/rtx_independent_proposal_replay_v2/offline.py` with extracted root argument	Full proposals, tapes, events and code; shared numerical routines; sanitized ordinal ordering
Original freeze objects	Each study's source/policy, prediction/tape, acknowledgement and first-live records; compact `INDEX.json` distinguishes content roles	Retained bytes and reported host/server ordering; Git/hash identity alone is not trusted time

sets cannot be rehashed by the offline reader. No functional HumanEval pass@k or online upload is implied.

P Post-Study Existing-Record Analyses

The major outcomes were known before these v0.8.6 calculations. Before recomputation, a written plan fixes the complete retained inputs, formulas, ordering and treatment of missing records. These are secondary descriptive or sensitivity analyses; no original endpoint, primary CI or bootstrap stream is replaced. Anonymous `derived/inputs.json`, `derived/results.json` and `analysis.py` allow standard-library replay.

P.1 B570 raw logits, margins and continuation

Each reserved run retains three raw target-logit rows for a common proposal-conditioned prefix, whether or not all proposals are reached by actual acceptance. Top-one/top-two are ranked over the full vocabulary before sampler transformations; margin is their score difference. A fixed proposal's raw logit is also retained (absent for the bonus row). These logits are neither normalized probabilities nor sampler-adjusted margins. Numerical fields change in all 72 runs; timestamps/metadata are excluded. The 216 rows and 195 reachable paired decisions are correlated, as are three repeats per prompt. Top-two ID changes occur in three rows; raw top-one IDs never change. Absolute top-one delta quartiles are 0.0388 and 0.1327, with p95 0.3437. Positive minimum raw margins coexist with unchanged immediate decisions; these summaries do not prove a whole-generation null effect or locate a numerical operation.

Table A14. B570 post-study raw-logit descriptors over all 216 paired rows. Δ = UB128 minus UB1; margins use retained top-two scores/IDs. Units are raw logits.

Statistic	Min	Median	Max
Δ top-one	-0.4275	-0.0443	2.3503
$\lvert\Delta\rvert$ top-one	0.0019	0.0695	2.3503
UB1 margin	0.2291	17.5021	33.6660
UB128 margin	0.2003	17.5128	33.5351

The new cohort is HumanEval/1, /8, /13, /21, /29, /36, /41, /51, /71, /74, /75, /79, /83, /84, /85, /92, /94, /110, /124, /131, /152, /153, /155, /163. Every primary and alternate endpoint reaches cap 128, with no earlier EOG. Source-before, restored and source-after serialized KV content hashes/lengths match at all 72 forks; consumed lengths and positions match. Recursive sampler cloning is source-audited, with matching recorded chain/seed/history. Six same-partition UB1-to-UB1 R0 restores have equal scores and full continuations; the complete R0 set has 24 existing executions including original/native equivalence. Sampler opaque-state bytes are not separately serialized. These controls support the sampled contrast, not identity of unavailable external binaries.

All sampled immediate decisions agree. Later UB128 target blocks persist and proposals evolve with the alternate accepted history. These secondary differences concern another

Table A15. All six B570 continuation-sensitive prompts. Zero-based generated positions, distance = first difference minus fork index. Every row is stable across all three repetitions, with 128/128-ID capped endpoints. Length-only divergence would occur at the shorter array's length; none occurs here.

HumanEval	Fork	First difference	Distance
/29	65	114	49
/75	65	80	15
/84	65	115	50
/110	65	84	19
/131	64	74	10
/163	66	97	31

Table A16. The complete M4 target-faster subset, from original medians. Ratios are relative latencies (×); lower is better. All six Guard runs fall back at round 8 but each Guard median remains above target-only. No transition component is inferred.

Task	Target (s)	Always/Target	Guard/Target
/559	11.9085	1.0270	1.0215
/95	10.9203	1.0150	1.0029

cohort; they cannot explain old H1's four divergences or a single-block restoration.

P.2 M4 opportunity between two fixed modes

From all 540 raw held-out observations, the three-condition medians exactly reproduce the original 60 prompt rows. Target-only is faster than Always on 2/60 prompts, both Spec-Bench; there are no equal Target/Always medians. Always/oracle GM is 1.0006919671×, where oracle is the per-prompt lower Target or Always median. It is limited hindsight choice between two measured fixed modes, not an arbitrary switching-policy lower bound. Guard/Always paired prompt ratios have min/median/max 0.9706/1.0022/1.3297 and quartiles 0.9985/1.0260. Family GMs are HumanEval 1.0099 and Spec-Bench 1.0446; these 30/30 families are descriptive, not stable small-stratum estimates.

A separate sensitivity stream uses Python Random seed 2026100101, 10,000 paired draws, sampling each family's 30 prompts with replacement in HumanEval then Spec-Bench order. Each draw computes GM directly; linear .025/.975 percentiles give Guard/Always [1.0135, 1.0423] and Always/oracle [1.0000, 1.0018]. Guard remains slower; original pooled primary and standalone secondary intervals stay unchanged. No CI is formed by dividing interval endpoints.

Table A17. Earlier two-path RTX cost study: post-study absolute prompt-median times (ms), 30 prompts. The original target/serialized GM and CI remain unchanged; the ratio of these distribution medians is not that GM.

Condition	Min	Q25	Median	Q75
Target	295.2	1186.9	1907.9	2730.3
Serialized	560.3	1821.1	2923.3	4053.6

P.3 358H conditional zero-draft accounting

For each path, sum matched baseline time B, complete target work T, draft D_t, residual X_t, committed IDs N and rounds L over all 90 processed observations. $T_b = B/N$, $C = N/L$, and $R_{\text{target}} = T/(LT_b)$. At the same progress, target work and baseline, with nonnegative draft/residual,

$$S = \frac{C}{D_{\text{real}} + R_{\text{target}} + X} \leq \frac{C}{R_{\text{target}}} = \frac{B}{T}.$$

Both paths retain 11,520 committed IDs; rounds are 4,677 CPU and 4,686 NPU. Full-precision processed times yield bounds CPU 0.8274529834 and NPU 0.8351858524×. Even removing both draft and residual costs cannot reach parity under these fixed conditions. This is arithmetic total-time accounting, not a bound on prompt-GM speedup, a measured optimization, separable replay saving, persistence gain or a bound after changing target execution/proposals.

P.4 Earlier RTX two-path absolute latency (30 prompts)

All 180 original condition times are milliseconds; medians retain three repetitions per prompt and the original 30-prompt pairing. The following distributions summarize 30 condition medians, not 180 independent prompts. Maxima are 2939.7 and 4490.6 ms. The JSON output also reports all 90 individual times per condition separately. Independent primary replay is 0.6458157742698285; the original report's one-ulp 0.6458157742698286 and original CI/seed remain recorded. The post-prefill clock boundary and measured hook probe in Section N apply; 0.6 μs p99 is about 0.0012% of the shortest 49.9572-ms R0 interval. That is a clock-call probe, not all logging overhead. The complete native-log files have median line counts 3516.5 target and 4353.5 serialized, and median byte lengths 266908 and 339335. Committed-ID event counts match for all 90 pairs. File-wide counts include initialization outside the timer; they do not measure charged logging cost or imply cancellation. No production-speed estimate is made.

Q RTX V4 Same-Set Prediction and Matched Cost

The joint V4 study has two arms: A tests complete-trajectory prediction; B tests reference qualification and instrumented cost. They use the same 19 tasks, but every B observation is newly executed. A has 285 formal conditions (228 pre-live and 57 live); B has 171. The 456 conditions are not independent samples. Every named comparison has 57 pairs and 19 distinct prompts.

Q.1 Exposure, failed V3 and capture repair

From the 164 HumanEval tasks, 145 exposed or ambiguous tasks are conservatively excluded. This includes 138 previously excluded tasks and seven newly exposed during V3: /2, /16, /59, /86, /157, /158 and /161. All 19 remaining tasks are selected; no replacements or outcome-selected subset are used. SHA256 ordering with salt `RTX_TRAJECTORY_SUFFICIENCY_V4|reserved|20261002|` gives /143, /31, /25, /81, /15, /141, /142, /18, /136, /99, /148, /140, /87, /76, /26, /49, /40, /109 and /66. This is a finite exposure-conditioned cohort, not a random representative sample. No-known-exposure is limited to the audited repository, records and accessible archives.

V3 retains its terminal failures. Of 288 planned A pre-live conditions, 72 are valid, one fails capture and 215 are not reached; none of 72 A live conditions executes. Of 216 planned B conditions, 54 are valid, one fails and 161 are not reached. The 126 valid prefixes and two failed records are not pooled into V4. Both capture failures occur on /59: native execution exits successfully and emits 128 committed IDs ending in EOG ID 151645, then reports limit followed by EOS. The old parser demanded one stop marker. V4 is a new protocol and cohort; regression checks accept the complete simultaneous EOG/cap boundary while retaining V3's original failed status. Missing, repeated, reversed or contradictory markers remain rejected.

The pinned vocabulary API identifies six EOG IDs without a decode. Captures require a contiguous complete committed-ID stream, agreeing generation count, first-inclusive EOG or cap128 and consistent terminal markers. Engineering gates for reference, replay and the three timer conditions each pass 36/36. Repair-specific gates each pass 3/3 and include 21 actual simultaneous EOG/cap conditions. Twenty observer pairs, 15 raw/protocol/statistical negative cases and 14 damaged endpoint fixtures are retained; engineering conditions total 430 and do not enter formal denominators.

Q.2 Configuration, input contracts and ordering

A uses Qwen3-8B/0.6B Q8_0, pinned llama.cpp b10839, CUDA0 with target 37/37 and draft 29/29 layers offloaded, context4096, logical batch2048, F16 K/V, one sequence, 24 threads and FA off. Processes and contexts are fresh. Target sampling is greedy with seed123; draft proposals are limited to two and use the pinned native sampler behavior. The target-only competitor has UB2 and no draft. H2 uses separate draft P and target R processes; fresh R_ONLY loads only target and sequentially reads H2's sealed tape. Native batched live uses its own drafting and accepts no tape. Numerical libraries and kernels remain shared.

R accepts a prompt at initialization, then only the allowlisted round, position, pending ID and proposals. It checks these against its own committed prefix and computes target sampling, acceptance, correction/bonus and rollback locally. Past-prefix assertions and tape integrity checks are not supplied future live answers. H2's proposals depend on its own earlier feedback, so tape construction is not draft-free. Application/API input separation and engineering rejection witnesses are retained; they are not adversarial OS isolation or an independent numerical implementation.

F0 fixes implementation, assets, effective settings, exposure, selection and gates before engineering; F1 fixes the predefined candidates before A pre-live. F2 seals all four sets of 57 arrays and full proposal tapes, with complete remote retrieval before native live. B has a separately sealed start state, timer and balanced schedule before all 171 fresh observations. Reported host/server records and Git membership support the preserved ordering, without independently trusted absolute process timestamps.

Completion uses a disclosed fan/host-policy amendment after predictor capture. The changed host policy is a study limitation, not an independently proved physically irrelevant adjustment. The original strict-host stop and predictor records remain unchanged. The author authorized continuation before any native live outcome; amended authority, complete predictions and tapes were sealed and retrieved before live. Numerical code, candidates, assets, gates, endpoints and cost schedule were not retuned. The same 228 predictor observations are reused, not a newly unexposed replication. The original stop and amendment remain in the full evidence packages.

Q.3 Full-array results and distinct failure cases

The main-text trajectory table gives A's primary counts. All five conditions are repeat-stable. H2/live, R_ONLY/live and H2/R_ONLY each agree 57/57; UB2/live and UB1 reference/live each agree 51/57. The latter two comparators disagree on different prompts. Both replay programs predict the two reference-divergent trajectories. Complete agreement supports empirical sufficiency for this finite path/cohort; UB2's failure rejects only that candidate. It proves neither necessity nor minimality of history and identifies no unique numerical cause.

Table A18. RTX V4 first differences, stable across three repetitions. Indices are zero-based generated-ID positions; replay matches live throughout. Equal aggregate UB1/UB2 counts hide different failures.

Comparator	Task	Index	Own ID	Live ID
Target-only UB2	/25	4	271	1406
Target-only UB2	/15	15	4199	9147
UB1 reference	/15	15	4199	9147
UB1 reference	/18	29	1191	3007

Table A19. RTX V4 matched qualification and cost: 171 fresh runs on 19 prompts × three × three paths. Target/serialized use UB1, batching UB2; speculation uses $k = 2$. Speedup = target/candidate latency. Only the blocked batched row has null speedup/CI. Serialization is faster on 0/19 prompts.

Path	ID gate	Speedup [95% CI]
Target	Reference	1.000
Batched	51/57 blocked	—
Serialized	57/57 qualified	0.677 [0.662, 0.692]

The compact index retains all 285 A arrays with per-condition endpoints, including the first EOG; it also retains all 171 B arrays and native timer fields. An export/recalculation after known outcomes is not a new prospective prediction freeze. Source mappings bind exported fields to original sealed prediction and record bytes; full capsules retain proposals, logs, source and process ledgers.

Q.4 Matched three-path cost and estimator

B uses one frozen instrumented frontend/core and the same recorded CUDA backend DLLs for target-only UB1, native batched $k = 2$ UB2 and serialized $k = 2$ UB1. The global native micro-batch option also propagates to draft; this is a comparison of whole execution configurations, not an isolated target intervention. Model, prompt, context, endpoint, logging, timer and schedule are matched. All 171 runs are fresh, although A has already exposed these tasks; B is a cost study, not another unseen prediction cohort.

Schedule seed2026100203 shuffles nine complete sets of the six condition permutations, plus three predetermined cyclic rotations. Across 57 prompt/repetition blocks, each condition occupies each order position 19 times. Qualification requires all 57 reference/candidate full arrays and stable reference repeats. Batched matches 51/57 and is blocked; serialized matches 57/57 and qualifies. Blocked speedup, CI and prompt ratios remain null. The main-text cost table does not infer a fast batched alternative. The same gate/estimate is retained here for the historical cross-reference.

For each prompt i and condition c, let m_{ic} be the median of its three native QPC intervals. For qualified serialization,

$$r_i = m_{i,\text{target}}/m_{i,\text{serialized}},$$
$$G = \exp\left(\frac{1}{19}\sum_i \log r_i\right).$$

The frozen 10,000 bootstrap draws resample the empirical distribution of paired prompt rows in the stated task order with Python Random seed2026100204; linear .025/.975 sorted percentiles give the CI. Original report $G = 0.6773510334393157$, CI [0.6616533074169813, 0.6918118724220992]; independent raw recomputation gives the same GM and CI [0.6616533074169811, 0.6918118724220993]. Floating-point differences are below 10^{-14}; neither result replaces the preserved report. There are 0/19 ratios above one. Reciprocal GM latency is 1.476339×; no CI is formed by dividing unrelated interval endpoints. The resampling is conditional on the selected exposure-audited cohort, observed condition medians and frozen configuration. It does not model exposure selection, the full HumanEval population or new sessions. The earlier 30-prompt 0.646 estimate remains a separate study, not evidence of an optimization trend.

QPC begins after synchronized initial target/draft prefill and before per-request sampler/state work, ending after final committed output, synchronization, response/log and slot release. Loading, initial prefill and teardown are excluded. A second chrono interval checks each QPC interval within $\max(0.020\text{ms}, 0.001T)$; the APIs share the Windows oscillator. Empty clock-call probes do not bound all logging. Observer controls check IDs, not logging-free latency. Cost applies to the recorded instrumented request boundary; no throughput, energy or unrestricted deployment estimate follows. Both reference and speculative B conditions contain 48 earlier-EOG and nine capped runs.

Q.5 Post-study V4 absolute latency descriptors

The v0.8.8 plan is written after all original outcomes are known, before this new calculation. The exact same 171 B QPC intervals are grouped into three-run medians per prompt and condition. Min, Q25, median, Q75 and max summarize 19 prompt medians per condition using linear order-statistic interpolation. Units below are seconds. These are empirical observation-spread descriptors, not confidence intervals. The paired primary GM and CI above remain unchanged; dividing the displayed distribution medians would produce a different estimator. No speedup is formed for blocked batching. Anonymous `derived/RTX_V4_PROMPT_MEDIANS_V088.csv` retains all 57 prompt-condition medians; `derived/GLOBAL_ANALYSIS_V088.json` retains full precision and cross-failure cases. `global_analysis.py` replays

Table A20. RTX V4 absolute prompt-median post-prefill latency (s): 19 medians per condition, each from three runs. Post-study spread, not uncertainty. Batched times are diagnostic; these distribution medians do not estimate the paired speedup.

Condition	Min	Q25	Median	Q75	Max
Target	0.532	1.441	1.607	2.396	2.920
Batched	0.641	1.730	1.910	2.953	3.248
Serialized	0.843	2.074	2.332	3.611	4.102

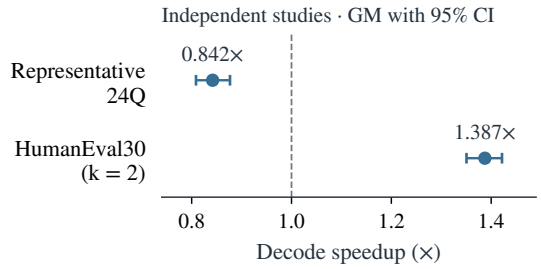


Figure 2. **Opposite profitability regimes on the same tested M4 stack**. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both pass the recorded re-tokenized-text comparator (72/72 and 90/90 pairs); these are not direct generated-ID comparisons or functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

them without models. A separate author-side NumPy computation from retained original B timer records cross-checks all 171 durations and five descriptors per condition; it is a CPU recalculation, not a new runtime replication.

Q.6 Exact anonymous entries and independent audit

The compact `anonymous-evidence-index-v0.8.8.zip` supports `python3 -B verify.py` without models or network. It retains the previous studies and post-study inputs, adds V4 arrays/times, and checks gates, failures, sample units, the frozen paired bootstrap and v0.8.8 descriptors. It does not contain raw KV/sampler state bytes and cannot rehash omitted binaries or authenticate original absolute chronology.

Full V4 A anonymous archive SHA256 is `befd1569 92138902624406a2c8c6f8854494caad7c7c 00df3c4cf3de5bd6f6c7`, length 89,274,896 bytes; B is `7be6bd011a2f3741d6788f19768a1f2f 5e40deceebb394ec9dd631a951fe2157`, length 116,890,414 bytes. The compact `INDEX.json` identifies each parts index, full digest, byte length and role. Concatenate numbered parts, verify the full hash, extract, and run the included `frozen_source/scripts/platform/rtx_trajectory_cost_v4/offline.py` with the extracted root. The two final anonymous capsules include the original CPU verifier and engineering/failure evidence. The separately indexed V2 H2 capsule remains unchanged.

An independent author-side CPU audit retrieved and checked all nine V4 engineering/pre-live/final author and anonymous capsules at the exact delivered Git tree, then ran each original offline entry. A separate parser reconstructed all 57 native A arrays and 171 B arrays/QPC intervals from raw logs and recomputed gates and the paired bootstrap. No model execution occurred. The readable receipt and byte-to-source mappings are preserved in the private evidence ledger. This is evidence verification, not an independently reproduced GPU run; sanitized derivatives retain the stated chronology and omitted-asset limits.

R Study Map and Historical Timing

The main text now centers the prospective predictions and system decisions. This section preserves earlier timing evidence and the opposite M4 workload outcomes without pooling them with later direct-ID studies. Historical M4/RTX/B570 timings use parsed-text comparators (H), not direct-ID qualification. Later Q gates do not certify those observations.

The representative frozen M4 sample gives 0.842× [0.809, 0.877], with 72/72 text-path-equal pairs and 1/24 prompts faster. Separate preregistered HumanEval fixed-$k = 2$ gives 1.387× [1.351, 1.422], 90/90 text-path-equal pairs and 30/30 prompts faster (Figure 2). The workloads and sessions differ: neither is a workload-only causal effect or general M4 profitability. Early MLX failures do not negate the historical Q8_0 Metal result.

Primary fixed-$k = 2$ HumanEval package-efficiency is 1.013× [0.990, 1.034]; the interval includes parity. The registered $p_{\min} = 0.9$ energy result remains secondary. The preregistered corrected synthetic ceiling was not executed, and the earlier diagnostic does not replace it.

RTX serialized CPU-to-GPU is 0.404× [0.390, 0.418], same-GPU drafting 0.650× [0.629, 0.673], both 90/90 H pairs and zero of 30 prompts faster. These controls retain different placement/integration costs, not an isolated transfer-cause estimate. B570 serialized timing is diagnostic, 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster; its formal failure/stop rule and board-energy boundary remain unchanged. All newer studies retain their own

Table A21. Studies supporting the main argument. Counts use prompts as sampling units; repeated runs are correlated. Historical campaigns remain separate in Appendix R. Prediction and qualification have different comparators.

Study	Question / comparator	Prompt unit	Endpoint / cost boundary
B570 H1	Frozen target-only prediction vs live; own UB128 reference	Reserved 24 × 3	Complete 128-ID arrays; no timing
B570 forks	Matched state, UB1 vs UB128 sampled block; persistent continuation	Separate 24 × 3	Scores, immediate decisions, later IDs
RTX V4 A	Predefined UB2 and replay vs native batched live	New 19 × 3	Four pre-live conditions + native live
RTX V4 B	Own UB1 reference vs batched/serialized, with matched cost	Same 19 × 3	Three fresh conditions; instrumented CLI
358H Phase 3	Own reference vs CPU/NPU draft interface	30 × 3	Direct-ID gate; complete loop
M4 runtime	Target, Always, Guard; two-fixed-mode hindsight oracle	Held-out 60 × 3	Direct-ID gate; charged loop

Table A22. Earlier campaign estimates, with original prompt GM/95% CI and original comparator. H = historical text path, Q = direct IDs. A fresh Q gate does not certify H timing. Counts and faster-prompt totals are stated in the text; V4 is a separate study in Section Q.

Path / workload	Speedup [95% CI]	Gate
M4 representative	0.842 [0.809, 0.877]	H
M4 HumanEval	1.387 [1.351, 1.422]	H
CPU to RTX	0.404 [0.390, 0.418]	H
RTX to itself	0.650 [0.629, 0.673]	H
B570 serialized	0.3087 [0.2927, 0.3246]	H*
358H CPU draft	0.734 [0.693, 0.771]	Q
358H NPU draft	0.352 [0.333, 0.369]	Q

*B570 timing remains diagnostic on a mismatching path. RTX denotes RTX 5070 Ti Laptop.

reference and cohort.